A formal bridge from two refutations of the Exponential Time Hypothesis to a standard multitape Turing machine*

Carlos Toledo[a,*]

[a]*Independent researcher, the cert-machine project, Brazil*

Abstract

In October 2026 OpenAI and Anthropic each published a Lean 4 proof that the Exponential Time Hypothesis (ETH) is false. Each statement is about a machine and an input encoding of its laboratory's making: Anthropic's `ETHIsFalse` is the negation of $s_3 > 0$ with s_3 taken over Mathlib's multi-stack Turing machines (`FinTM2`) reading a custom encoding of duplicate-free formulas over the variables $x_0, \ldots, x_{n-1}$; OpenAI's `exact_count_and_sat` is a statement about its own algorithm on the multitape machines of its ParityLifts library. The step from either to a standard model is left to prose. We prove in Lean that `ETHIsFalse` implies ETH is false as stated in the cslib-community library `complexitylib`: for every $\delta > 0$ a deterministic multitape Turing machine decides 3-SAT on the plain binary encoding of every formula within $c \cdot 2^{\delta n} \cdot (L+1)^e + c$ steps. We then port both proofs into the same Lean, Anthropic's (809 files, 47 edited for the newer toolchain) and OpenAI's together with Daniel Hawkley's two theorems that carry it to `ETHIsFalse` (3,210 files, 20 edited), so that on each side the proof, the statement and the bridge are checked in one environment and no step crosses a version boundary. Every step was checked from pinned bytes by Lean's kernel, by nanoda (77,276 and 101,618 declarations under the two chains), by `leanchecker --fresh`, and by Lean's judge `lake comparator`: with six checkers on Anthropic's chain; on OpenAI's chain, whose export is

*v0.3 of 2026-11-06, built from repository commit 4f47947207be. Changes in v0.3 (2026-11-06): `con-leche` and `con-ron` from their master branches accept both exports of OpenAI's chain, and the exports are deposited (Sections 4 and 5, Data availability). Changes in v0.2 (2026-11-05): OpenAI's chain added, its proof and Hawkley's two links ported into the same Lean (Sections 2–4); the judge's four bundled checkers stopping at their own limits on that chain's export, and the re-run with named checkers that the verdict reads, written out (Sections 4 and 5); the open item on OpenAI's version boundary replaced; the complexitylib pull requests. v0.1 (Anthropic's chain only) is archived in release v2026.11.05, doi:10.5281/zenodo.23966840; v0.2 in release v2026.11.05b, doi:10.5281/zenodo.23985003; this version is archived as release v2026.11.06, doi:10.5281/zenodo.23992217. Every measured number is a macro read at build time from the records named in the Data availability statement; a record that no longer supports a sentence stops the build.

*Corresponding author.

Email address: `carlos@carlostoledo.co` (Carlos Toledo)

2.45 GB, the judge's four bundled external checkers stop at their own resource limits as shipped, and the verdict reads the judge re-run with named checkers (Lean's paranoid kernel, our pinned nanoda, lean4lean with its fuel raised, Lean's default kernel), which also accepts OpenAI's own challenge file against the ported proof. 12 planted wrong variants were refused. Whether the formal statement is the textbook ETH remains a reading of a definition. The proofs were written by AI agents directed by the author, who also wrote the checker; this is not an independent review.

Keywords: Exponential Time Hypothesis, 3-SAT, formal proof, Lean, machine models, independent checking

---

1. What is claimed, and what is not

Two AI laboratories published Lean proofs in October 2026 that the Exponential Time Hypothesis of Impagliazzo and Paturi (2001) is false (Anthropic, 2026; OpenAI, 2026). Both proofs are accepted by Lean's kernel, which we confirmed independently for both, against their own statements, with a second kernel (Toledo, 2026). The question that readers raised is not whether the proofs check but what they state: each statement is about a machine and an input encoding of its laboratory's making, and the step from there to a standard machine is argued in prose.

This note closes that step for both statements. *Claimed:* a Lean theorem, checked as described in Section 4, that Anthropic's `ETHIsFalse` implies the negation of ETH as defined over cslib's deterministic multitape Turing machine with a binary encoding of arbitrary formulas; a second Lean theorem in which Anthropic's own proof of `ETHIsFalse`, ported, supplies the hypothesis in the same environment; and a third in which OpenAI's own proof, ported together with the two theorems of Hawkley (2026) that carry it to `ETHIsFalse`, supplies the hypothesis in the same environment. *Not claimed:* that ETH is false; the correctness or novelty of either laboratory's algorithm outside its Lean development; that a formal definition is the textbook one, which no machine decides (Section 5). v0.1 of this note covered Anthropic's chain only; the title footnote lists what v0.2 adds.

2. The statements

Anthropic's. `MainResults.lean` at commit 16a5ce41 (Lean v4.33.1, sha256 `47f78a9550d95b37…`) defines `InTime 3` δ: some `FinTM2` machine with finite alphabets and a constant c decide, for every n and every 3-CNF φ on $x_0, \ldots, x_{n-1}$ without repeated clauses, the satisfiability of φ within $\lfloor c\, 2^{\delta n} \rfloor$ steps, reading `encClauses 3` φ. Then $s_3 = \inf\{\delta : \texttt{InTime 3}\ \delta\}$, `ETH` is $0 < s_3$, and `ETHIsFalse` is its negation.

OpenAI's. The theorem OpenAI's release checks is `exact_count_and_sat` of `OAI.Algorithms.SubexponentialSAT.SqrtLogSpace` (family 378 of `openai/math`, commit 7ef5c0cc, Lean v4.34.1): a statement about its own algorithm, on the multitape machines of its ParityLifts library and that library's encoding `encodeFormula`. The step to Anthropic's statement is Hawkley's (Hawkley, 2026), in two theorems written inside OpenAI's Lean: `not_positiveRateETH : ¬ PositiveRateETH`, derived from OpenAI's `original_sat_main`, where `PositiveRateETH` is the ETH predicate of the ParityLifts library; and `not_positiveRateETH_implies_ethIsFalse : ¬ PositiveRateETH → ETHIsFalse`, which runs ParityLifts' machines on `FinTM2` within a linear number of steps and converts `encClauses` into `encodeFormula` within a polynomial of one fixed degree. We had replayed both with two kernels inside OpenAI's Lean (Toledo, 2026); until v0.1 the copy of `ETHIsFalse` there and the one the bridge starts from were only read to be the same proposition, across Lean v4.34.1 and v4.36.0-rc2.

complexitylib's. `ETH` is $\exists\, \delta > 0$, $\neg$ `DecidableInExpTime (KSAT 3) encCNF numVars` δ, where `DecidableInExpTime` asks for one deterministic multitape Turing machine (cslib's `Turing.MultiTapeTM`) and an exponent e such that the machine decides membership in `KSAT 3` within $O(2^{\delta n}\,(L+1)^e)$ steps on the binary encoding `encCNF` of length L, for every formula: variable labels are arbitrary natural numbers, and literals and clauses may repeat. This is the form of the hypothesis on a standard machine model with a reasonable binary encoding; its distance from the textbook definition is discussed in Section 5.

3. The bridge, and the two chains

```
EthBridge.Anthropic.not_eth_of_ethIsFalse : MainResults.ETHIsFalse
→ ¬ ComplexityAbstracts.SubexponentialSATAbstract.ETH
```

Given $\delta > 0$, take $m = \lceil 2/\delta \rceil + 1$ and a `FinTM2` decider at a rate below $1/m$. The proof has four parts. (i) *One step of any program in degree one.* complexitylib already simulates a `TM2` program on the multitape machine in polynomial time, but the degree of that polynomial depends on the program, and here the program is chosen after δ, so the bound would be circular. We prove that a single step of any `TM2` program is computable in degree one, uniformly in the program (`DegreeStep.stepTotalD`). (ii) *The run on a unary clock.* The t-step run is a degree-two fold over the pair (input, t in unary) (`DegreeRun.runBitD`), tied to Mathlib's `TM2OutputsInTime` (`runBit_eq`). (iii) *The reduction.* A polynomial-time map from `encCNF` to `encClauses 3` relabels the variables densely, removes repeated clauses, and passes n along, so that the decider can write $2^{n/m+c_0}$ in unary and run that many steps (`Words.prep_polyTime`,

`Prep.prep_spec`: satisfiability and duplicate-freeness preserved). (iv) *The rate.* The composed decider runs within $2^{2n/m} \cdot \mathrm{poly}(L)$ steps, and $2/m \leq \delta$ (`Assemble.decidableInExpTime`).

Anthropic's chain. The bridge elaborates Anthropic's `MainResults.lean`, byte for byte, against complexitylib's Lean (v4.36.0-rc2) rather than Anthropic's (v4.33.1). To remove that version boundary from the chain, Anthropic's proof (the import closure of its `Solution.lean`, 809 files) was ported to the same toolchain: 762 files are byte-identical; 47 were edited for library renames, missing imports and tactic repairs, each edit listed with both digests in the port's ledger. No edit touches `MainResults.lean`, and every statement changed only by a forced rename is listed. Then

```
EthBridge.AnthropicChain.not_eth : ¬ETH :=
not_eth_of_ethIsFalse MainResults.ethIsFalse
```

is checked in one Lean. The ported proof is a modified copy; since Lean checks whatever proof is supplied against the unchanged statement, the edits affect no decided fact, but the proof checked is not Anthropic's bytes.

OpenAI's chain. The same was done on OpenAI's side. The import closure of Hawkley's two bridge files, 3,210 files (3,183 of OpenAI's, byte-equal to `openai/math` at 7ef5c0cc; 27 of Hawkley's, at `thinkthoughts/math` d6b79b7d), was ported to the same toolchain: 3,190 files are byte-identical to their pins and 20 were edited for the newer Lean and Mathlib, each edit listed with both digests in the port's ledger. Hawkley's import of `MainResults` is redirected to the vendored copy, so that his second theorem proves the same `ETHIsFalse` the bridge consumes; his two statements, `MainResults.lean` and OpenAI's own challenge file are unchanged. Then

```
EthBridge.OpenAIChain.not_eth : ¬ETH :=
not_eth_of_ethIsFalse (not_positiveRateETH_implies_ethIsFalse
not_positiveRateETH)
```

is checked in one Lean. The same remark applies: the proof checked is a modified copy of OpenAI's and Hawkley's bytes, against statements that did not change.

4. How it was checked

Each trusted statement was written by hand and pinned by digest before any proof was decided (bridge `c84eae828cc11769…`, Anthropic's chain `229e7202c5a00303…`, OpenAI's chain `dcce9dcb1ce6d1c6…`). On rented Ubuntu machines, from the published commits (claimbox 235fbfa, c1442b1 and bdf357e, on complexitylib 7c66eb71), each run (a) checked the statement block byte for byte against the pin, `MainResults.lean` against

Anthropic's published digest and, for a ported proof, every file against its pin or against the port's ledger (byte-equal, or modified exactly as declared, nothing else); (b) built from source; (c) printed the axioms from our own file: `propext`, `Classical.choice` and `Quot.sound` only, for the theorem and its helpers; (d) exported the theorem with `lean4export` and replayed it in nanoda, a kernel written independently of Lean's, which accepted 23,430 declarations under the bridge, 77,276 under Anthropic's chain and 101,618 under OpenAI's; (e) ran `leanchecker --fresh`; (f) ran Lean's judge `lake comparator` against a challenge written on the machine from the pin. Three planted variants per theorem had to be refused, and were: the statement closed by `sorry` (shown by the axioms file and refused by nanoda), the statement through a new axiom (likewise), and the statement silently changed (refused by the statement check and rejected by the judge for a mismatch). The judge's sandbox could not start on the rented containers; the judge ran without it, and the records say so.

The judge on the two chains. On the bridge and on Anthropic's chain the judge ran as shipped, `--paranoid`, and Lean's kernel, `leanchecker-paranoid`, `lean4lean`, nanoda, `con-leche` and `con-ron` all accepted (the last two over 22,058 and 74,801 declarations; the chain's export is 0.65 GB). On OpenAI's chain the export is 2.45 GB and the judge as shipped did not finish: Lean's paranoid and default kernels accept, and each of the four bundled external checkers stops at a limit of its own, in its own words. The bundled nanoda 0.4.17 overflows a worker thread's stack, and that build has no stack setting; `lean4lean` stops with "(kernel) deterministic timeout", its fuel counter reaching zero; `con-leche` and `con-ron` stop with "internal error: fuel exhausted: whnfCore", a compiled constant, which their charter calls a decline, never a verdict. None is the rejection of a declaration. Each stop was reproduced by hand on the comparator's own export, and the verdict reads the judge re-run with named external kernels: `leanchecker-paranoid`, our pinned nanoda (`ammkrn/nanoda_lib` 8de7e0c4, 16 threads, a 1 GiB thread stack; it accepts the export in 142 s), `lean4lean` with every fuel field raised 100× (it accepts in 1,382 s), and Lean's default kernel. That run accepts the chain (77 min), rejects the planted statement on the statement, and accepts OpenAI's own challenge file `ComparatorChallenges/SubexponentialSAT.lean`, unchanged (sha256 `4183b2057cff350d…`; its module name in the judge's configuration is the one declared change), against the ported proof of `exact_count_and_sat` in the new Lean (73 min). The record keeps the shipped run beside the named one, and the amendment is dated in the pre-registration. `con-leche` and `con-ron` are not among the checkers that decided OpenAI's chain. After the record, on their maintainer's request, `con-leche` (master 7cb7a45c and dc8ca857) and `con-ron` (master 97eead2d), built from source and run by hand on the same export, accept all 98,686 declarations, and all 93,373 of

the export of OpenAI's own challenge; `con-leche` needs 65.9 GiB at 16 workers and 51.4 GiB at 8. Their fuel stops are fixed upstream, not yet in a Lean release, and these runs are outside the decision. Every stop of the four bundled checkers was therefore a resource limit or an old build, and none the rejection of a declaration.

Beside the chains. Two further results use the same checks. The library's own proof that its `ETH` is false (commit 1b7d62a8; nanoda over 49,025 declarations) is a re-proof on the standard machine, not a bridge from either laboratory's statement. And `EthBridge.WordRAM.not_eth_ram` refutes the word-RAM form of the hypothesis, with $O(\log L)$-bit words, from complexitylib's native model counter rather than from either laboratory's statement (nanoda over 49,735 declarations, six checkers): complexitylib's simulator bounds simulated addresses by the running time, which rules out logarithmic words for super-polynomial runs; our variant bounds them by the work space, and the model counter runs in polynomial space. In the records the word-RAM theorem is C1 and the bridge of Section 3 is C2. C1 and the reusable part of C2 are offered upstream: complexitylib pull requests #2509 and #2510 (5 November 2026, on the maintainer's invitation in #739), approved by the library's reviewer agent, which is not human approval; merge pending.

5. What stays open

(1) *The textbook match is a reading.* That complexitylib's `ETH` is the hypothesis of Impagliazzo and Paturi (2001) is a reading of a definition: the machine, the binary encoding, the parameter n and the polynomial factor in L are standard choices, and any further distance is argued, not proved. (2) *OpenAI's chain was decided by the judge with a shorter list of checkers.* As shipped, `lake comparator --paranoid` did not finish on OpenAI's chain: its four bundled external checkers stopped at their own resource limits on the 2.45 GB export (Section 4), and the verdict reads the judge re-run with named kernels, four where Anthropic's chain had six. Two of the six, `con-leche` and `con-ron`, finish this export only in builds from their master branches, which accept it (Section 4); the full list returns with a Lean release that bundles them. (3) *The ported proofs are modified copies* (Section 3). (4) *The judge ran without its sandbox*, which protects the judge from a hostile build; the builds were the author's. (5) *One author on both sides.* The statements were written by the author; the proofs, the ports and the corollaries were written by AI agents (Claude, Anthropic) under the author's direction; the checks were run by the author's tooling. Roles are separated, but the same person is on both sides, and no second party has replayed the records; doing so is the next step, and the Data availability

statement gives what it needs: for OpenAI's chain, a 239 MB download and one checker on a 64 GB machine.

Data availability

The Lean sources are in `https://github.com/carlostoledo1891/claimbox/tree/bdf357e/lean/eth-bridge` (the ports' ledgers are `AnthropicEth/PORT.md` and `OpenAIEth/PORT.md`). The records, the pinned statements, the machine scripts and every fact of every run are in `https://github.com/carlostoledo1891/cert-machine`: `certs/eth-chain.json`, `eth-openai-chain.json`, `eth-c2.json`, `eth-c1.json`, `eth-comparator.json`, `eth-cslib-d1.json`, `eth-hawkley-heavy.json`, `eth-hawkley-bridge.json`; `corpus/eth-refutation/`; `tools/runpod-eth-chain.sh` and `runpod-eth-openai-chain.sh`; the pre-registration of OpenAI's chain with its dated amendments, `notes/eth-openai-chain-preregistration-2026-11-05.md`. The two kernel exports of OpenAI's chain, with the commands that rebuild them, are deposited as doi:10.5281/zenodo.23991348 (pinned in `corpus/eth-refutation/openai-chain/exports-deposit.json`); the master runs are under `corpus/eth-refutation/openai-chain/runs/`. A summary with the figure is at `https://carlostoledo.co/reports/eth-chain.html`.

Declaration of generative AI use

The Lean proofs, the ports and parts of the tooling were written by AI agents (Claude, Anthropic) directed by the author, who wrote the statements, chose the checks and takes responsibility for the claims. One of the two laboratories whose claims are discussed is Anthropic.

References

Anthropic, 2026. formal-math: `eth/`, a Lean formalization that the Exponential Time Hypothesis is false. `https://github.com/anthropics/formal-math`, commit 16a5ce41.

Hawkley, D., 2026. OpenAI–Anthropic ETH cross-formalization (RPI-ETH-010). `https://formalize.now/378/`; the Lean sources in `https://github.com/thinkthoughts/math`, commit d6b79b7d.

Impagliazzo, R., Paturi, R., 2001. On the complexity of k-SAT. J. Comput. Syst. Sci. 62, 367–375.

OpenAI, 2026. The Exponential Time Hypothesis is false; and a Lean formalization in `https://github.com/openai/math`, commit 7ef5c0cc.

Toledo, C., 2026. Two claimed refutations of ETH, checked. `https://carlostoledo.co/eth/`.